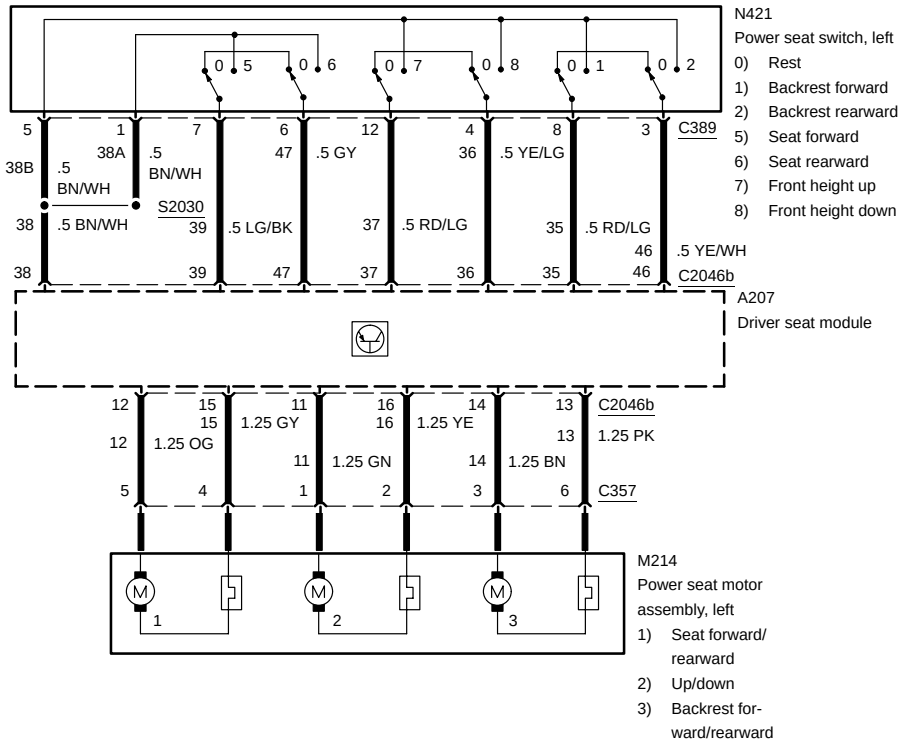with memory -3